

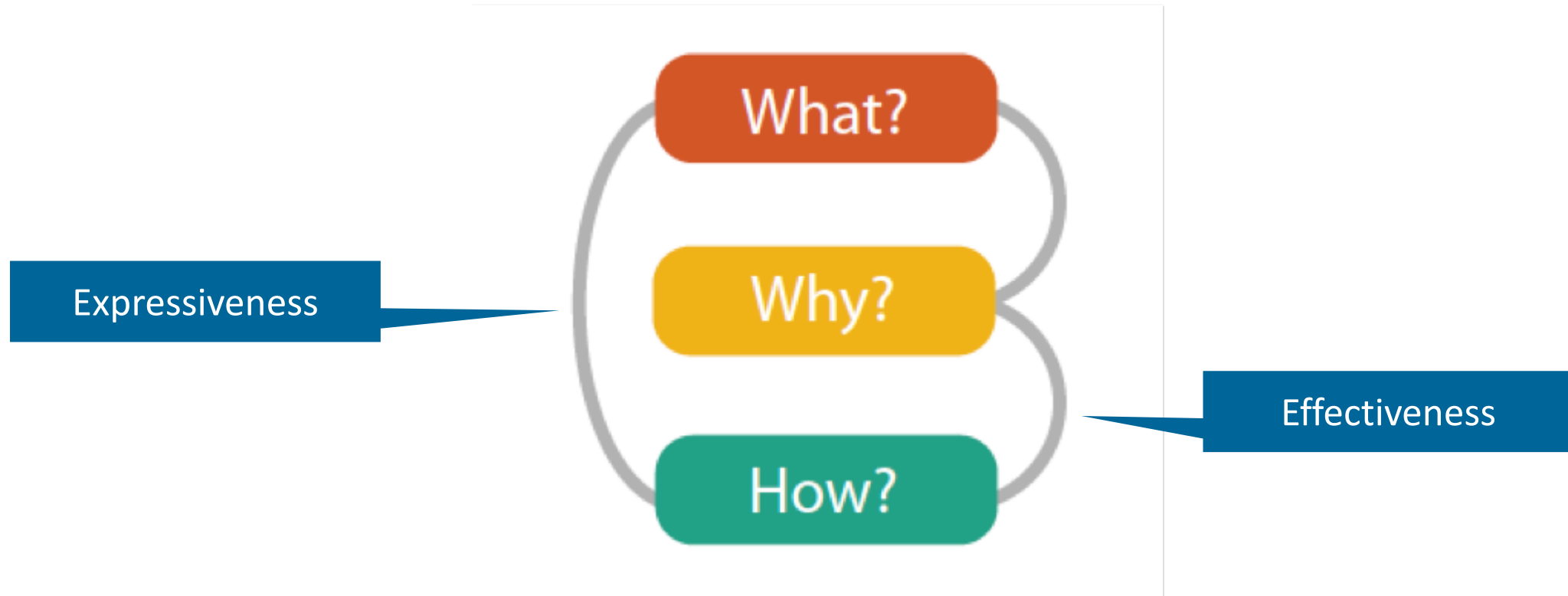
VU Information Visualization (193.171)

Q&A

Examples

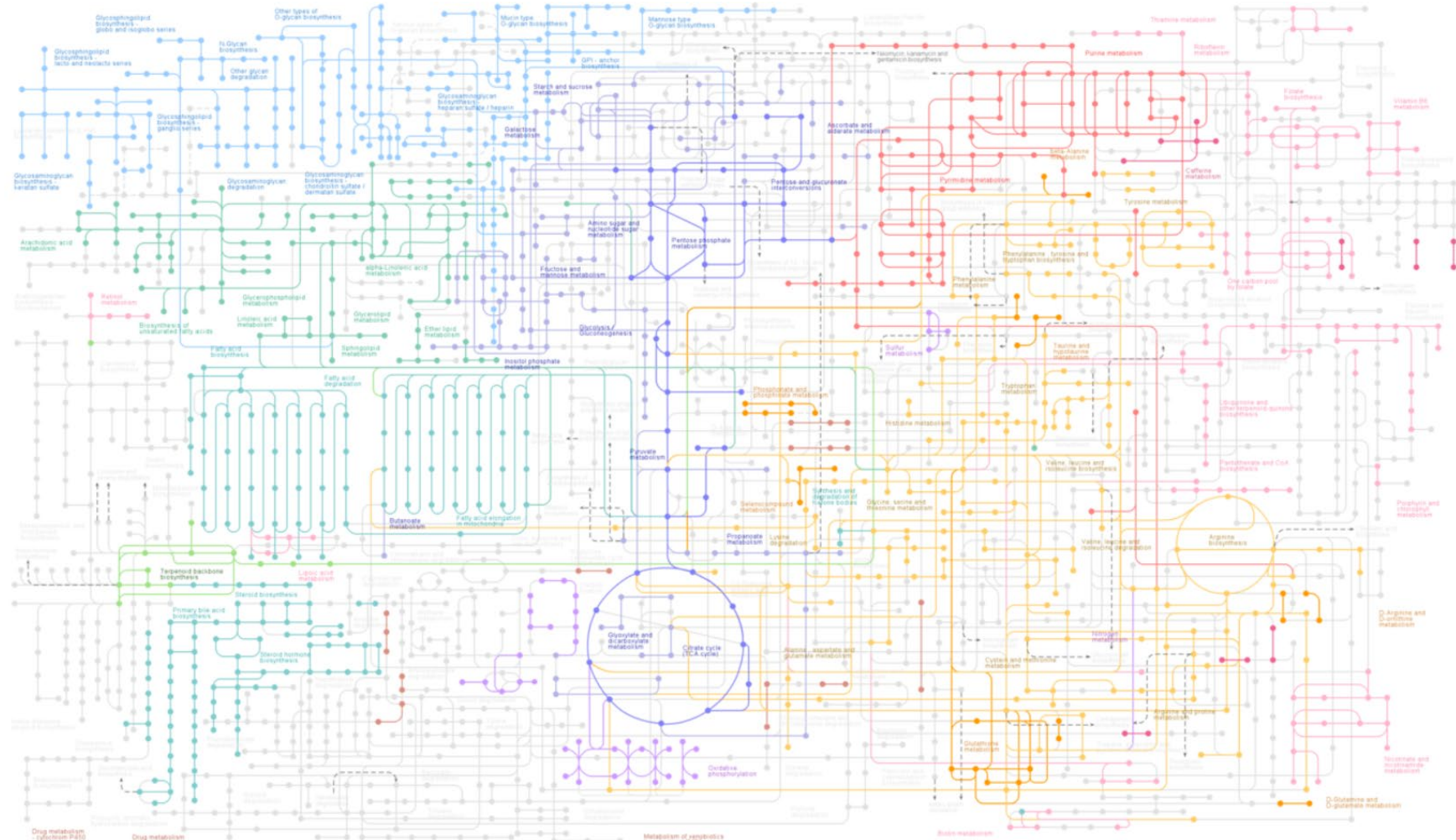
Manuela Waldner





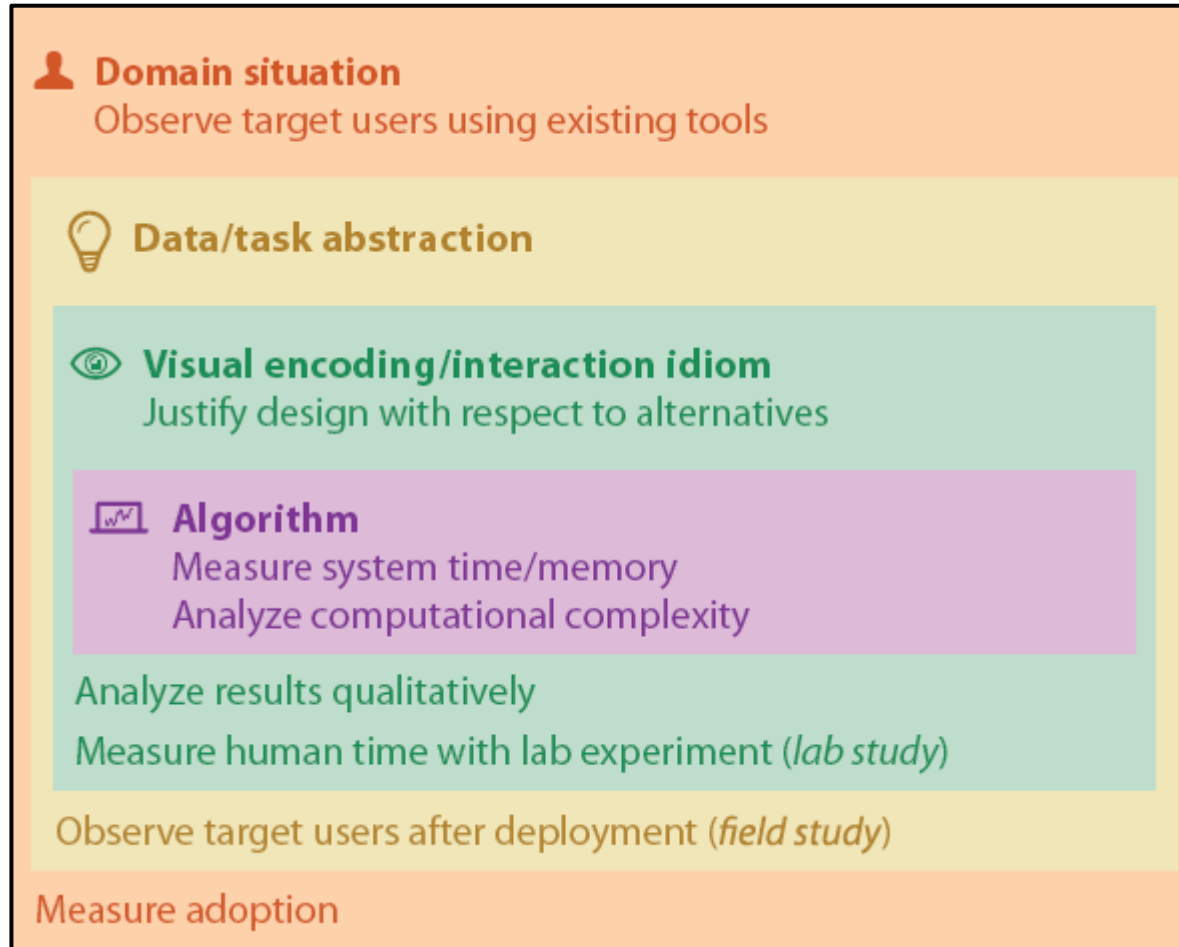
Visualization Challenges

- Data size
- Data complexity



https://www.genome.jp/kegg-bin/show_pathway?map01100





Exam



- 24.6.2026 – (De-)register through TUWEL until **22.6.2026 (noon)!**
 - Slot 1: 12:00 – 13:00
 - FAV 01 (+ InfoVis VO students)
 - If required: EI 8
 - Slot 2: 13:00 – 14:00
 - FAV 01
 - EI 8
- **Final assignment will be published on 23.6.2026!**



- 24.6.2026 – (De-)register through **TISS** until **17.6.2026!**
- 12:00 – 13:00 in FAV 01



- **17.9.2026** – registration via TUWEL (VU) or TISS (VO) - soon!
 - Last attempt for VU!
- VO: further exam dates upon request (most likely oral)



■ Participation:

- Only guaranteed participation for assigned slot
- Be on time (s.t.!)
- Student (picture) ID required!

■ Procedure

- Maximum 45 minutes
- Closed book (no materials, electronic devices, or conversations)



■ VU

- Lectures 5-11
- Foundations need to be known and understood!
- Mandatory readings

■ VO

- Lectures 1-11
- Mandatory readings
- Mandatory readings VO



- VO
 - Participation in exam → certificate
 - No exam = no certificate
- VU
 - Everybody who submitted something (e.g., Exercise 0) will be graded
 - Participation in regular exam → certificate in July
 - Everybody else → certificate in September / October
- UE / VU Project + Exercise / VU Project
 - Certificate in July



- Different course settings = different maximum points!
- Grades based on % of maximum points:

Grade letters		S1	U2	B3	G4	N5
Lower boundary in %	?	88.00	75.00	63.00	50.00	0.00
New lower boundary in %	?	88.00	75.00	63.00	50.00	0.00

- Multiple exam attempts = multiple certificates:
new certificates overwrite existing ones!



- Lecture (40)
 - Intermediate TUWEL quiz (10) *
 - Online user study participation (2)
 - **Final written exam (28) ***
- Exercises (25)
 - Visualization literacy test (2)
 - Python notebook (5)
 - D3 interactive linked views (18)
- Data analysis project (35)
 - Project proposal (5)
 - Project implementation + documentation (25)
 - Final presentation (5)

50% per block required for
positive grade!



If you have not filled out the course selector,
you will be graded as a VU student!



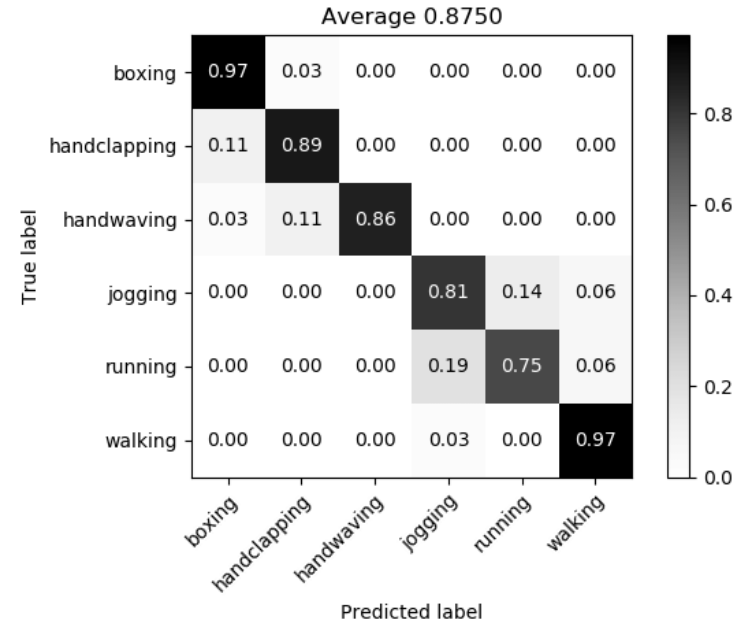
Exam Topics

Example Questions



-

■ Example:



- ✓ ■ Boxing is never confused with handwaving
- Jogging is most difficult to classify



- Describe
 - Sketch
 - Multiple choice questions
 - Properties
 - Hyperparameters
- Recursive subdivision (Mosaic plot, treemap)
 - Just noticeable difference
 - Scagnostics
 - Class consistency
 - PCA
 - Multi-dimensional scaling
 - T-SNE
 - UMAP
 - [Hexagon] binning
 - Kernel density estimation
 - Multi-dimensional tables / data cubes
 - Parallel aggregation
 - Feature vectors (e.g., from neuron activations)
 - Local perturbations
 - Partial dependence
 - Counterfactual explanation
 - Force-directed graph drawing
 - Layered graph drawing
 - Graph morphing
 - Foresighted graph layout
 - ...



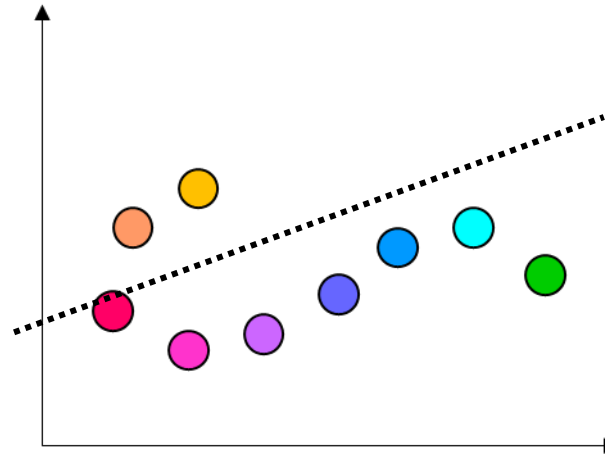
■ Example:

■ Select the correct statement(s):

- ✓ ■ Counterfactuals represent the minimal set of changes to feature values so that the prediction changes to a different output.
- Counterfactuals represent the maximum change of feature values so that the prediction switches to the opposite output.
- There is only one possible counterfactual explanation.
- ✓ ■ Counterfactuals should be chosen so that they are plausible.



- Example: sketch 1D PCA projection



- Given data and task, suggest and justify suitable
 - Data processing
 - Visualization design
 - Interaction design
 - Justify against alternative(s)
- Multivariate data
- Unstructured data
- Multidimensional data
- Tall data
- Large graphs
- Set-typed data
- Dynamic graphs
- ML model
- ...



- Given is a dataset about traffic accidents in Europe from 1970 to 2020. The **tabular data** has the following format:
 - Accident ID | longitude | latitude | year
 - The table has more than **10 million rows**. There are no missing data.
- How would you generate an **interactive visualization** (e.g., for the web) so that users can explore the distribution of accidents **spatially and temporally**?
 - Note: the visualization does not need to be multi-scale. Visualization on a coarse spatial level is sufficient.
- How would you **preprocess** the data?



- Preprocessing: we have a **scalability** challenge!
 - Binning into pre-defined regions and years → perceptual scalability
 - Data cube / multi-dimensional table → interactive scalability
- Visualization: two **linked** views
 - Spatial (regions can be brushed)
 - Choropleth map
 - (Hexagonal) bin map
 - Sequential color map
 - Temporal (individual years or year ranges can be brushed)
 - Line chart or histogram



- Point deductions:
 - Not identifying or handling of scalability challenge
 - No linked views → only spatial or temporal analysis possible
 - Ineffective processing (e.g., KDE)
 - Only spatial or temporal binning
 - Minor:
 - Missing names for visual encodings
 - Not mentioning binning or aggregation (just explaining it)
 - Not describing / naming interaction design
 - Not mentioning color mapping (or non-optimal color mapping)

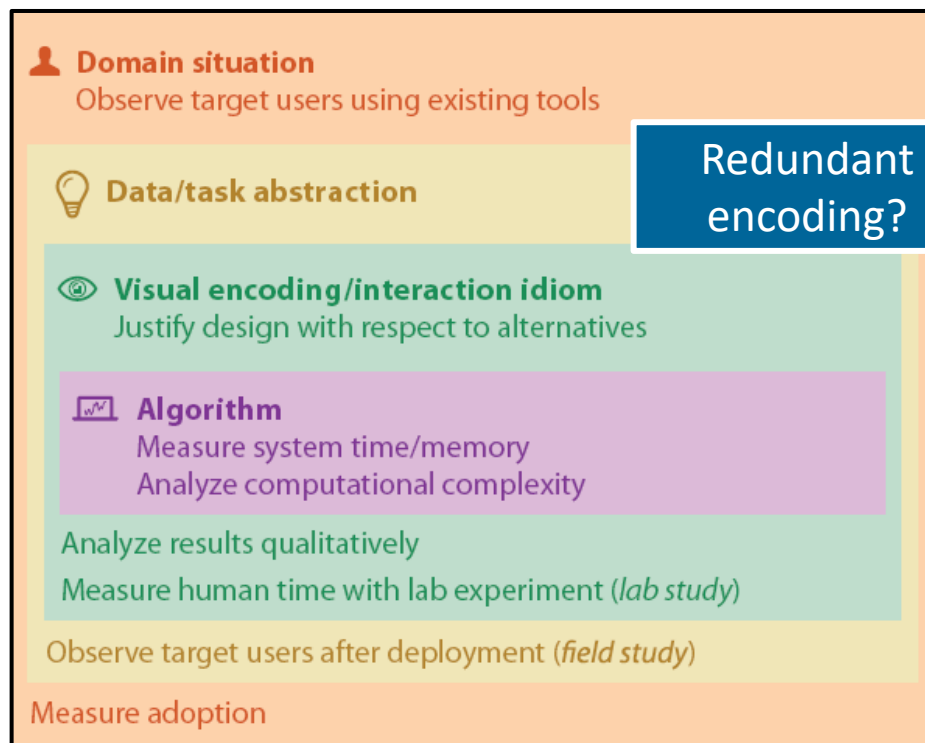


- Given a topic in visualization research, describe how you would validate your project

- Example (from an oral exam → too elaborate for written exam):
 - There are different options how to visually encode data items of different **classes (or sets) in static scatterplots**. How would you evaluate the effectiveness of possible visual encodings? Describe the design of your planned validation approach.



- Evaluation of visual encoding
 - Design alternatives



➔ Identity Channels: **Categorical** Attributes

Spatial region



Color hue



Motion



Shape



Spatial channel
occupied

Categorical
colors and
shapes possible
for static
display

Marks as Links

➔ Containment



➔ Connection



Link marks
possible for
showing class
association



- Hypotheses
 - Color will be more effective than shape [identity channel ranking]
 - Connectedness will be more effective than color [Gestalt Law]
 - Redundant encoding will be more effective than single-channel encoding [channel interference]
- Independent variable(s)
 - Visual encoding [color, shape, connection, color + shape, color + connection, shape + connection, color + shape + connection]
- Control / random / additional independent variables?
 - How many classes?
 - Balancing of classes?
 - How many points?
 - Distribution of points?
 - Chart size?

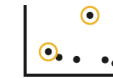


■ How to measure “effectiveness”?

■ Why do we visualize?

■ What are our low-level actions?

→ Identify



→ Compare



→ Summarize

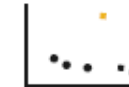


■ What are the data aspects of interest?

→ Trends



→ Outliers



→ Features



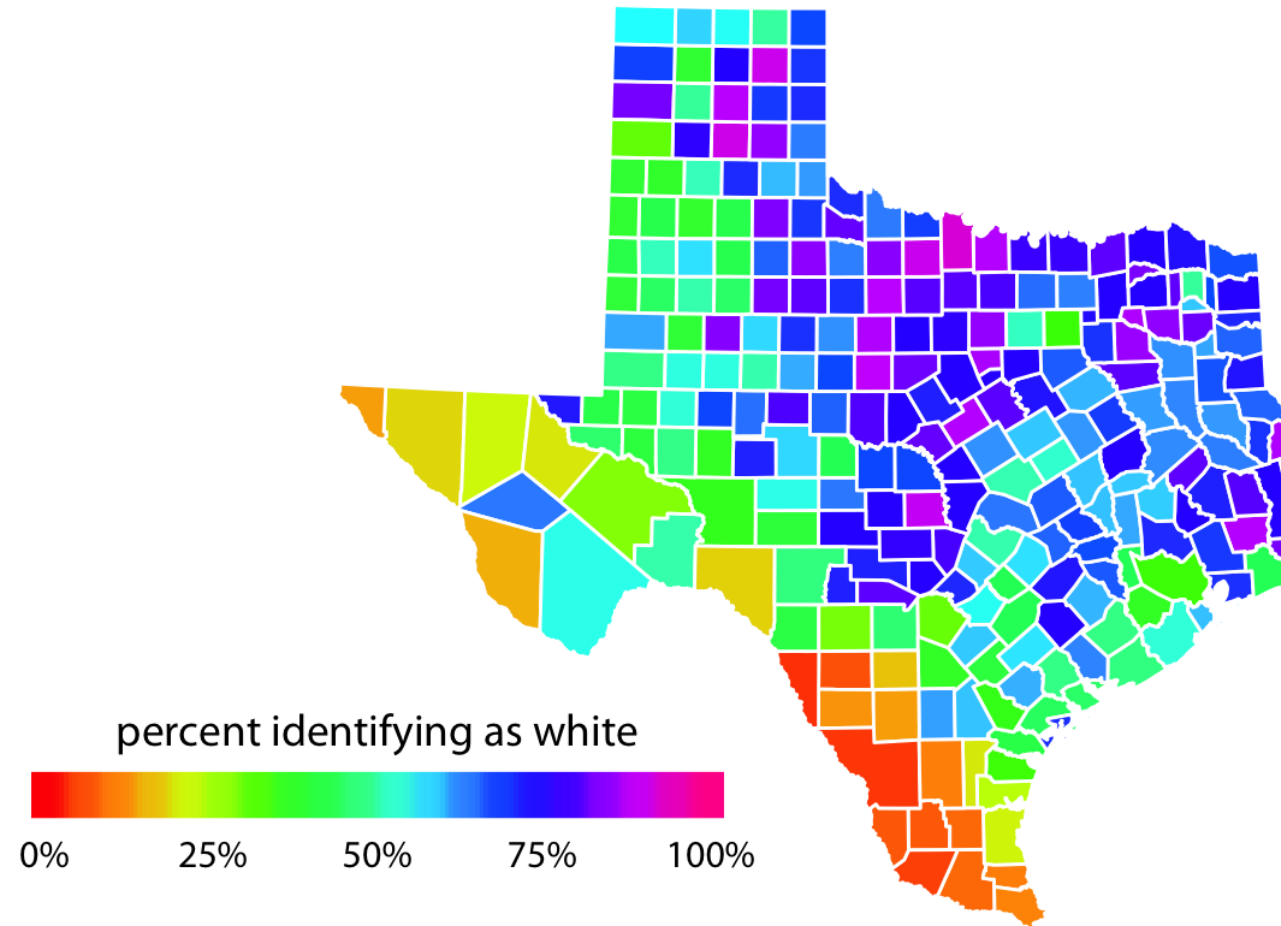
- How to measure “effectiveness”?
 - Task: click on all items that belong to a given class
 - Dependent variable(s)
 - Accuracy (false positives, false negatives)
 - Time?
 - Any confounding variables?
 - Crosses may be harder to select than circles



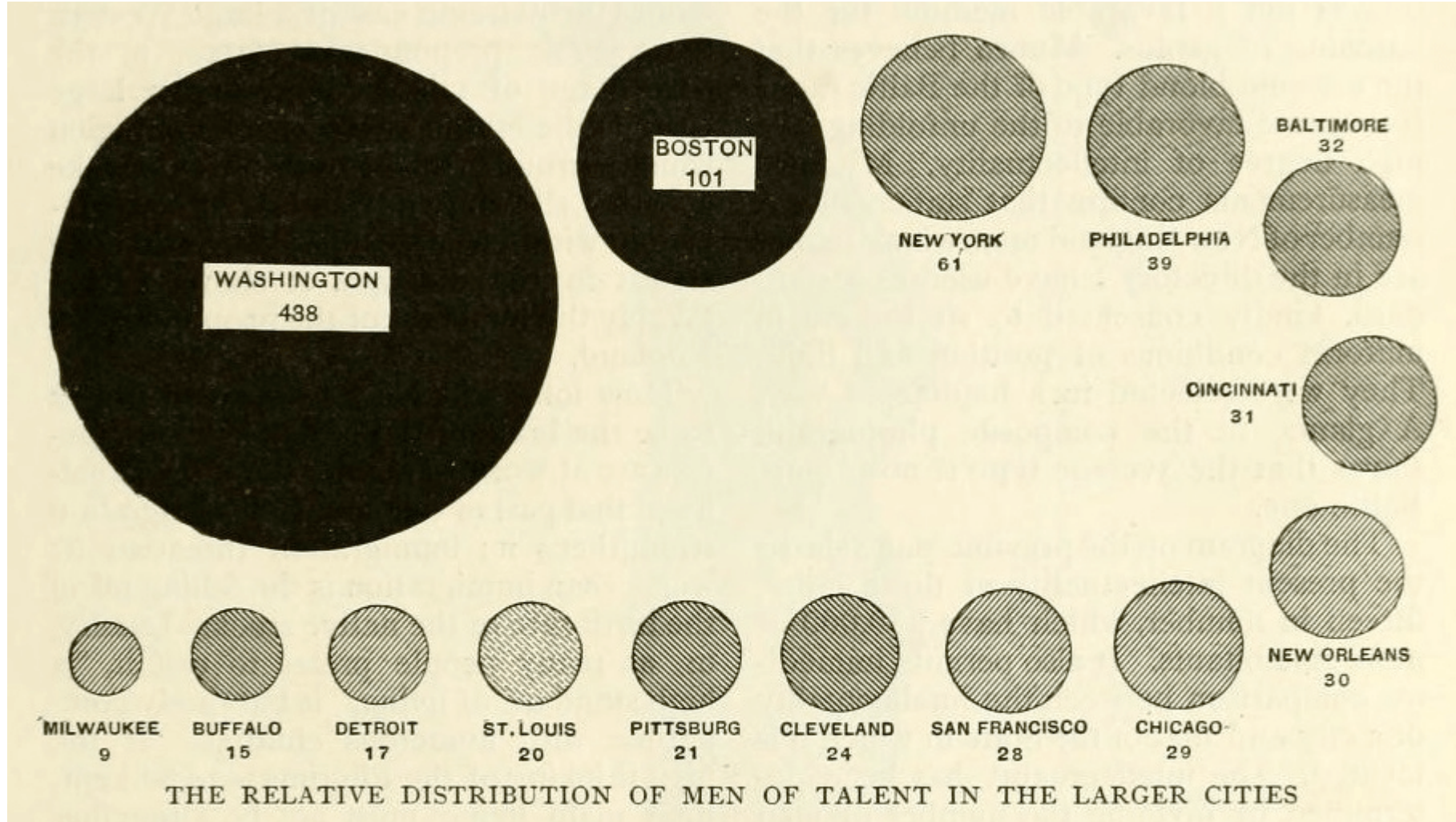
- Given a visualization:
 - Discuss strengths and potential problems, for instance based on
 - Data-ink ratio
 - Data aspects of interest and effectiveness
 - (Hidden) data transformations
 - Common design principles (e.g., 3D, “lie factor”)
 - Ordering (e.g., bars, axes)
 - Expressiveness & color maps
 - Effectiveness (e.g., Stevens’ power law)
 - Channel interference
 - Suggest alternative visual encodings



■ Example:

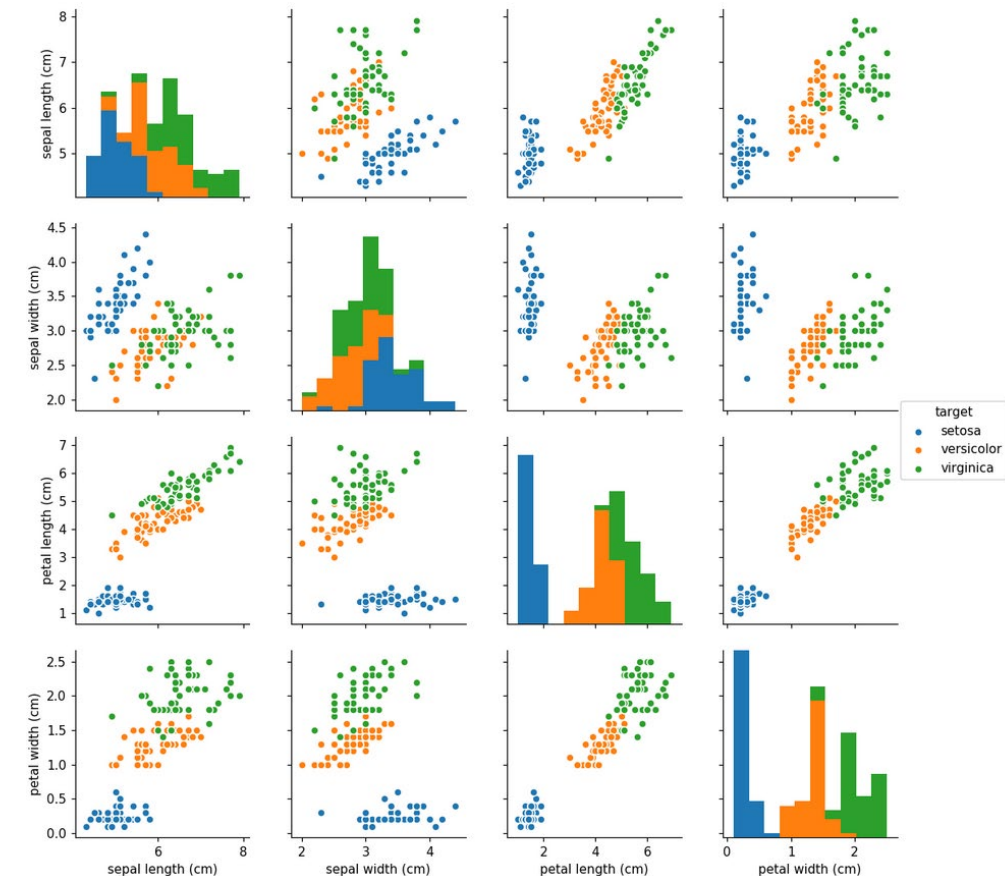
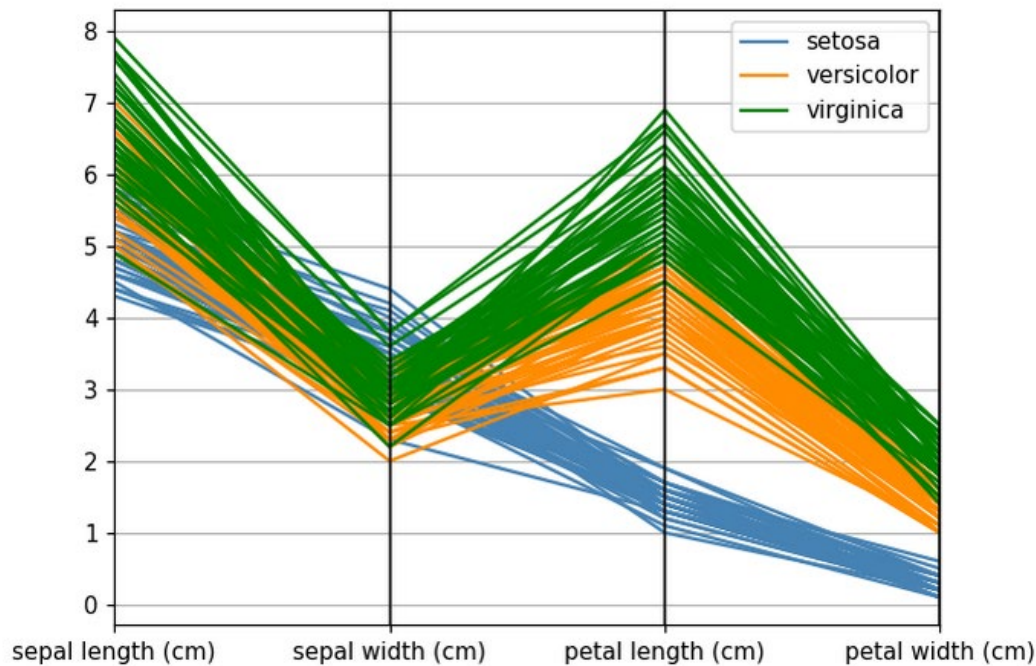


■ Example:



Alternatives

- Given two (or more) visualizations for the same data, discuss the advantages and disadvantages of the options
- Example:



- Transfer from one representation to another (sketch the solution)
- Example:
 - *Given:* [0, 5, 18, 22, 24, 34, 37, 38, 38, 39, 40, 41, 46, 49, 50, 58, 61, 66, 78, 85, 98]
 - *Sketch:* histogram, box plot



- Mark a data item highlighted in one visualization in the respective other visualization
- Examples:
 - Scatterplot $\leftrightarrow$ parallel coordinates
 - Parallel coordinates $\leftrightarrow$ star plot
 - Heatmap $\leftrightarrow$ grouped / stacked bar chart
 - Node-link diagram $\leftrightarrow$ adjacency matrix
 - Treemap $\leftrightarrow$ rectilinear node-link tree representation
 - ...



- List visualization-related concepts, for examples:
 - Raw data issues in data tables
 - Factors affecting scalability
 - Sources of latency
 - Aesthetics criteria for graph drawing
 - Simple solutions to overplotting that do not require any data preprocessing
 - Model-agnostic AI interpretability methods
 - Techniques to visualize set-typed data
 - Methods for validation of visual encoding and / or interaction design
 - Typical dependent variables in visualization user studies
 - ...



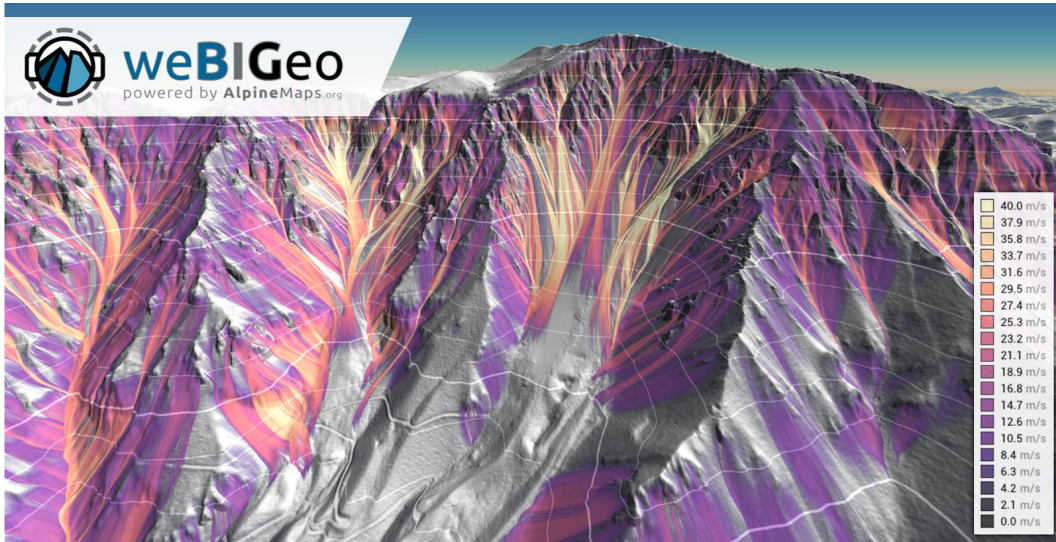
More Visualization!



- Visualization VU 6.0 EC (special topic lectures, GPU programming tutorials, [real-time] algorithm lectures, guest talks, interactive visualization project)
- Visual Data Science VU 3.0 EC (lectures about visual analytics, visualization environments / applications, data analysis exercises)

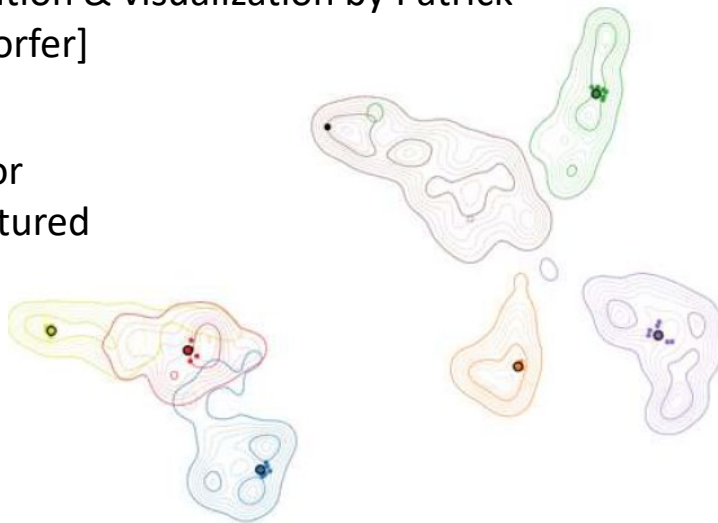


<https://www.cg.tuwien.ac.at/courses/Topics>

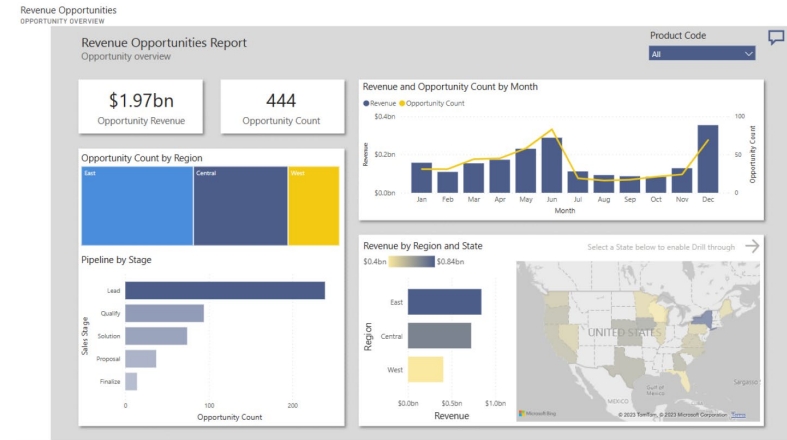


[Real-time avalanche simulation & visualization by Patrick Komon & Gerald Kimmersdorfer]

[Prototypical Networks for Visualizing Large Unstructured Data by Mario Stoff]

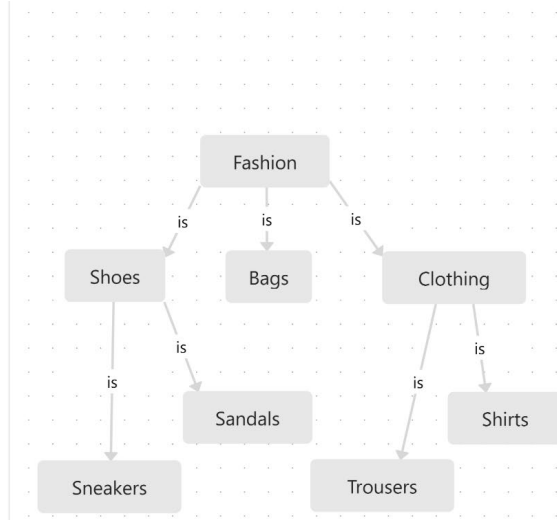
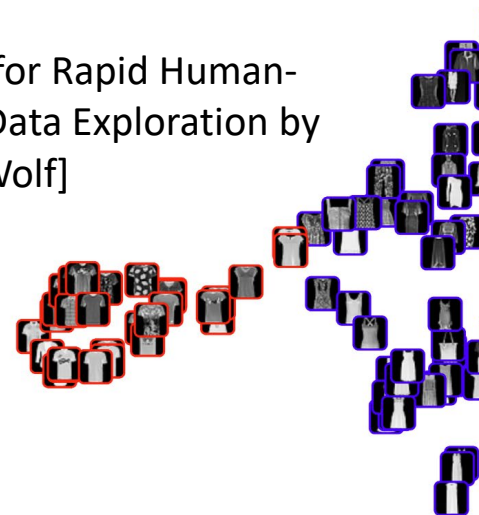


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[LLM-Enhanced Dashboards by Philipp Holzer]

[Sandbox for Rapid Human-Machine Data Exploration by Dominik Wolf]



Thank you!

Questions?

